

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Mechanical) (Credit System) Degree Examinations**

January - 2023

18MEE65 – NONDESTRUCTIVE TESTING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer Five full questions choosing Two full questions from **Unit – I & Unit – II** each and **One full question from Unit – III.***

Unit – I

	Marks	BT*	CO	PO
1. a) Give comparisons between Destructive Testing with NDT.	5	L*1	1	1
b) Outline the nine distinct areas of NDT and explain briefly any two.	5	L2	1	2
c) Explain the working principle of Eddy current inspection and with circuit diagram explain its stage by stage working.	10	L2	2	1
2. a) Sketch and explain Solvent removable liquid penetrant inspection.	5	L2	1	1
b) Describe the necessity of demagnetization after the MPI.	5	L2	1	2
c) Explain the basic principle of CT and with help of suitable sketches Distinguish CT with Radiographic inspections.	10	L2	2	1
3. a) Enumerate the magnetic field direction during circular & longitudinal Magnetization in MPI and Highlight advantages of MPI.	10	L2	1	1
b) List the applications and advantages of eddy current inspections	5	L4	1	1
c) Explain the basic principle of thermal inspection.	5	L2	2	1

Unit – II

4. a) Describe the B-scan setup for a basic pulse echo ultrasonic inspection system with neat sketch and highlight its applications and advantages.	10	L2	3	1
b) Enumerate the working principle of X-ray Radiography	5	L2	4	1
c) Write an explanatory note on radiation characteristics of gamma rays.	5	L1	4	2
5. a) With neat sketch illustrate the data presentation of C-scan setup for a basic pulse echo ultrasonic inspection system and list its applications.	10	L2	3	2
b) Explain briefly the factors affecting the selection of a radiographic film.	6	L2	4	1
c) Obtain the advantages and limitations of radiographic inspections.	4	L2	4	1

P.T.O.

6. a) Describe the general characteristics of Ultrasonic waves.	10	L3	3	1
b) Explain the basic principle of Real-Time Radiography.	5	L2	4	1
c) Briefly explain the applications of radiographic inspections in weldments.	5	L2	4	1

Unit – III

7. a) Explain the working principle of acoustic emission inspection	8	L2	5	1
b) Distinguish between Acoustic emission inspection with other NDE methods.	6	L4	5	1
c) Write note on Acoustic emission sensors and preamplifiers.	6	L2	5	1
8. a) Explain clearly the A E instrumentation principles and signal measurement parameters.	10	L2	5	1
b) Describe the detailed applications of A E in material studies, quality control and Structural & other studies.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
